Stochastic Process

Stochastic – This word means ‘probabilistic’ or ‘probability-based’ or ‘random’.

Process – It is a system that consists of a series of experiments.

Stochastic Process – A system of random experiments studied by a series of random variables.

Example 1: Two players are gambling using a fair coin. Player 1 has \$3 and Player 2 has \$2 as initial capitals. After each toss, the winner gets \$1 from the opponent. The game ends when one player has all the money. Let

$$X_n = \text{amount of money Player 1 has at "time" } n.$$

Then, $X_0 = 3$ (initial capital); $X_1 = 2$ or 4; $X_2 = 1$ or 3 or 5; $X_3 = 0$ or 2 or 4 or 5; and so on. These random variables together form a stochastic process.

- The random variables are NOT independent. So, they should be studied together.

Exercise 1: Consider Example 1. Calculate $P(X_3 = 2)$. [Hint: Use a tree-diagram. The result is 3/8.]

- Unconditional probabilities (like above) are difficult to calculate. Conditional probabilities are easy to calculate. For example: $P(X_3 = 2 \mid X_2 = 3) = 0.5$.

Discrete-time and continuous-time processes

If the system is observed at discrete time points $n = 0, 1, 2, \cdots$, we call it a discrete-time process. We write the process as $\{X_n;\ n = 0, 1, 2, \cdots\}$.

Example: See Example 1.

If the system is observed continuously over time t, we write the process as $\{X(t); t \geq 0\}$ and call it a continuous-time process.

Example: Number of customers that enter a Post Office in the interval $(0, t]$ is a continuous-time process.

States of a process

A value of a random variable of a stochastic process is called a state of the process.

Example: In Example 1 above, “2” is a state. $X_3 = 2$ means Player 1 has \$2 at time 3.

State space

The set S of all possible values of all variables is called the state space.

Example: In Example 1 above, $S = \{0, 1, 2, 3, 4, 5\}$ is the state space.

State space differs from <u>sample space</u>. A sample space is the set of all possible values of a <u>single</u> random variable. In Example 1, the sample space for X_2 is $\{1, 3, 5\}$.

R code practice:

Write an R function that takes any square matrix P as input and gives P^n as output, where n is any positive integer.

Types of stochastic processes:

1. Process with discrete time and discrete state space.
   Example: Consumer preference observed on a monthly basis.

2. Process with continuous time and discrete state space.
   Example: Number of customers that enter a Post Office in any interval of time.

3. Process with discrete time and continuous state space.
   Example: Amount of profit at the end of every month.

4. Process with continuous time and continuous state space.
   Example: Amount of profit in any interval of time.

Markov Process

Let $\{X_n;\ n = 0, 1, 2, \cdots\}$ be a discrete-time discrete-state stochastic process. If

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \cdots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i)$$

then the property is called Markov property and the process is called Markov process.

In a Markov process, if the present state of the system is known, the future is independent of the past.

Example: See Example 1. Here,

$P(X_3 = 4 \mid X_2 = 3, X_1 = 2) = P(X_3 = 4 \mid X_2 = 3)$

When X_2 is known, X_1 is not required to calculate the above probability.

Example (A process with no Markov property):

Scholarship in the 4th Year will be given based on history of scholarships in the past student life.

(When will it have Markov property?)

Markov Chain

A Markov process with discrete state space is called a Markov Chain.

(According to some authors, e.g., U. N. Bhat, a Markov process with discrete time is called Markov Chain.)

Example: The process in Example 1 (gambling) is a Markov Chain, because (i) it has Markov property and (ii) the state space is discrete. It is a discrete-time Markov Chain.

Example: (Customer arrival) Let $X(t) =$ Number of customers coming to a bank in $(0, t]$. Then, $X(t)$ has Markov property, because, for example, if $X(3.5)$ is known, $X(4)$ does not depend on $X(2.5)$. Here, $X(t)$ is a continuous-time Markov Chain.

Discrete-time Markov Chains

Transition probability

Let

$$p_{ij}^{(m,n)} = P(X_n = j \mid X_m = i); i \in S, j \in S$$

be the conditional probability that the process will have transition to state j at time n given that it was in state i at time m.

Example: (Gambling) If the coin is fair

$$p_{31}^{(2,4)} = P(X_4 = 1 \mid X_2 = 3) = \frac{1}{4}$$

When $n = m + 1$, we have

$$p_{ij}^{(m,m+1)} = P(X_{m+1} = j \mid X_m = i); i \in S, j \in S$$

which is called one-step transition probability at time m.

Time-homogeneous Markov Chain

When $p_{ij}^{(m,m+1)}$ is independent of m, we say that the Markov Chain is time-homogeneous. In that case, we drop m and write the one-step transition probability as $p_{ij}^{(1)} = p_{ij}$.

Also, the n-step transition probability for a time-homogeneous Markov Chain can be written as $p_{ij}^{(m,m+n)} = p_{ij}^{(n)}$. For $n \geq 2$, we cannot drop n.

Example: (Gambling)

$$p_{34}^{(2,3)} = P(X_3 = 4 \mid X_2 = 3) = \frac{1}{2}$$

$$p_{34}^{(4,5)} = P(X_5 = 4 \mid X_4 = 3) = \frac{1}{2}$$

Therefore, the chain is time-homogeneous. We can write both the probabilities as $p_{34}^{(1)} = p_{34}$.

Time-nonhomogeneous Markov Chain

Example:

Let us consider marital status of a man over discrete time, where time is age in years. Let

$S = \{\text{Unmarried, Married, Divorced, Widowed, Remarried}\}$.

$p_{UM}^{(21,22)} = 0.05$ (say)

$p_{UM}^{(26,27)} = 0.10$

$p_{UM}^{(31,32)} = 0.25$

We see that, transition probability changes over time.

Example:

Let $S = \{H, D\}$, where *H* means healthy and *D* means diseased.

$p_{HD}^{(19,20)} = 0.02$ (say)

$p_{HD}^{(60,61)} = 0.30$

We cannot write $p_{HD}^{(1)}$ because at 19 and 60, the probabilities of transition from healthy to disease are different.

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Transition probability matrix

Let a discrete-time Markov Chain with state space S be time-homogeneous. Let the one-step transition probabilities p_{ij}, $i \in S, j \in S$, be arranged in a matrix

$$P = (p_{ij})_{m \times m}$$

where m is the number of elements in S. The matrix P is called the transition probability matrix (TPM). It satisfies the following two conditions:

(i)

$$p_{ij} \geq 0 \text{ for all } i \in S, j \in S.$$

(ii)

$$\sum_{j \epsilon S} p_{ij} = 1 \text{ for all } i \in S.$$

Example: (Gambling) Here, $S = \{0, 1, 2, 3, 4, 5\}$. The 6×6 transition probability matrix P is as follows:

$$\begin{array}{c c} & \begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

Exercise (Food Stores): There are two food stores, A and B, in an area. Weekly data on customer preferences show that, 15% of customers of Store A go over to Store B next week, and 10% of customers of store B go over to Store A the next week. Construct the TPM.

Solution:

$$P = \begin{array}{c c} & \begin{array}{cc} A & B \end{array} \\ \begin{array}{c} A \\ B \end{array} & \begin{pmatrix} 0.85 & 0.15 \\ 0.10 & 0.90 \end{pmatrix} \end{array}$$

[Calculate P^n for $n = 2, 3, 4, \cdots$. Result for large n is interesting. R code for this problem is given later.]

Exercise: A telephone receptionist at a department store can process only one call at a time. She can be either 'idle' (0) or 'busy' (1). The following data were obtained by observing the receptionist's states after every 30 seconds for a period of 25 minutes:

1 10111 01011 10011 01011 11100 11110 10011 10110 10101 11010

Estimate the transition probabilities and construct the TPM.

Solution:

	0	1	
0	3	14	17
1	15	18	33
	----	----	50

$$\hat{P} = \begin{array}{c c} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} 3/17 & 14/17 \\ 15/33 & 18/33 \end{pmatrix} \end{array}$$

R code for matrix power

Exercise: Write an R function that takes any square matrix P as input and gives P^n as output, where n is any positive integer.

Solution:

The following code is written for beginners. It assumes that the inputs P (square matrix) and n (positive integer) are correctly provided by the user.

```
matpower = function(P, n)
  {
    Pnew = P
    if (n > 1)
      {
        for (i in 1:(n-1))
          {
            Pnew = Pnew %*% P
          }
      }
    Pnew
  }
```

We should now check whether the function written above works properly. Let us try the function with the transition probability matrix P obtained in the 'Food Stores' exercise. We can use the following R code:

```
P = matrix(c(0.85,0.10,0.15,0.90), 2)
matpower(P, 5)
matpower(P, 60)
```

Why are the rows in P^n identical for $n \geq 60$?